



Adaptive Modeling to Inform Environmental Management

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Abstract

Ecological modeling captures natural phenomena and processes, aiding in our understanding of ecology, conservation biogeography, and hydrology. The integration of ecological modeling into adaptive management frameworks is, however, underutilized. Adaptive modeling can enhance adaptive management by integrating new information and responding to changing environmental conditions, which facilitates dynamic and effective decision-making. This paper addresses challenges in implementing adaptive modeling, such as the need for collaborative model development, clear communication of model outcomes and limitations, and data and code accessibility. We propose a suite of potential actions to enhance communication, collaboration, and model utility. Focusing on water resource management, we demonstrate how adaptive modeling can inform management decisions and improve conservation outcomes. We highlight the importance of integrating adaptive modeling into future adaptive management plans to ensure models are relevant, reliable, and effectively utilized in resource management and conservation.

Introduction

Adaptive management provides a structured approach to managing ecological systems under uncertainty, emphasizing iteration, learning, and responsiveness. Within this context, ecological modeling, whether mathematical, computational, or conceptual, is a vital tool. Models inform key adaptive management functions such as goal-setting, forecasting, scenario evaluation, and monitoring (Walters and Green, 1997). They are fundamental to modern ecological science, supporting decision-making in fields such as conservation biogeography, water quality assessment, and hydrology (Guisan et al. 2013; Franklin 2013; Esselman

and Stevenson 2023; Olson and Cormier 2019; Grantham et al. 2022; Irving et al. 2018). These tools enable predictions of species distributions (Chee and Elith 2012), nutrient loading (Gal et al. 2009), and population and community dynamics (Mathwin et al. 2024; Peterson and Freeman 2016), particularly under conditions of environmental change (La Sorte and Jetz 2010; Walther et al. 2002). Their application in restoration planning, species management (Aiello-Lammens et al. 2011), environmental flow prescription (McKay et al. 2019), fish passage (Januchowski-Hartley et al. 2019), and risk assessment (Blair et al. 2022) continues to grow, with an expanding role anticipated in future planning (Weiskopf et al. 2022).

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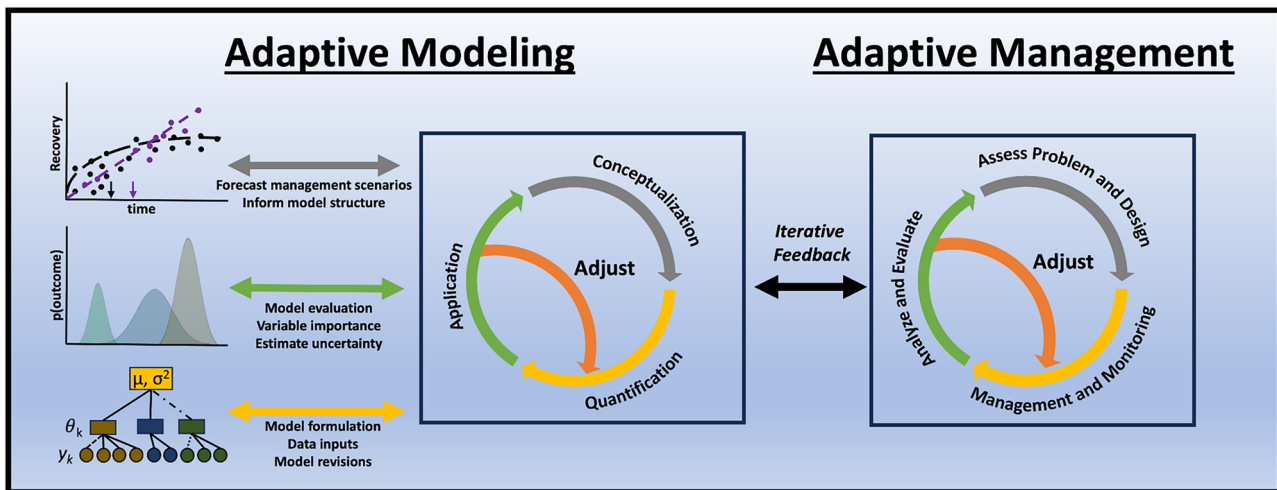


Fig. 1 Conceptual framework showing the integration of adaptive ecological modeling with the adaptive management cycle. The left panel illustrates the adaptive modeling cycle, in which models are iteratively conceptualized, parameterized, applied, and adjusted to improve model accuracy and precision relative to the management action. This process incorporates data inputs, model formulation and revision, and evaluation of model performance and uncertainty. The

right panel shows the adaptive management cycle, where problem assessment and design, management and monitoring, and analysis and evaluation inform iterative decision-making. Bidirectional feedback between modeling and management highlights how model outputs guide management actions, while monitoring data and management outcomes are used to update and improve models over time

Ecological models can play a vital role at each phase of the adaptive management cycle: assess problem and design, management and monitoring, analyze and evaluate, and adjustment phases (Convertino et al. 2013; Pahl-Wost 2007, Fig. 1). By integrating new information that describes changing environmental conditions, these models can facilitate dynamic and effective decision-making (e.g., Arhonditsis et al. 2019). Adaptive management occurs across a range of decision contexts, from ecosystem restoration to dam operations and wastewater discharge management (e.g., Dai et al. 2025, Failing et al. 2013, Littles et al. 2022). These systems involve different actors, including scientists, regulators, and managers, who shape how ecological models are developed and used. In such dynamic environments, models would be able to evolve alongside the management process. We therefore argue that modeling within adaptive management would also be adaptive - i.e., adaptive modeling.

The term 'adaptive modeling' has been defined as an approach where models are iteratively updated through time to incorporate new information as data are collected or understanding evolves (e.g., Lermusiaux 2007). Despite this definition, adaptive modeling is rarely implemented as a distinct, integrated process within the adaptive management cycle. Models are treated as static tools rather than dynamic frameworks to be updated and re-applied as new information becomes available. This creates a disconnect that limits the potential for models to support truly adaptive learning and iterative decision-making. The underutilization of actual adaptation that is often seen in

'adaptive models' may stem from challenges related to complexity and collaboration, such as weak stakeholder engagement, insufficient communication, and barriers to long-term model usability and trust. These challenges affect both modelers and non-modeler stakeholders - here defined to include decision makers, regulators, management and monitoring staff, community members, and other groups affected by or engaged in environmental decision-making that rely on model outputs. Bridging this gap requires better practices for incorporating modeling into ongoing management cycles (Fischenich et al. 2019; Peterson and Freeman 2016).

Early work examining the use of ecological models in environmental management emphasized the importance of iterative model development, stakeholder engagement, and trust-building when applying models to complex ecological systems (e.g., Schmolke et al. 2010; Smajgl 2010). More recent work has expanded these ideas through frameworks that integrate ecological knowledge, stakeholder engagement, and decision processes within ecosystem-based management (e.g., Langhans et al. 2019). Despite these advances, practical challenges remain in linking modeling processes with monitoring design, decision thresholds, and iterative management adjustments within adaptive management programs. Here, we build on this body of work by focusing specifically on how ecological models can function as adaptive components of management cycles, being updated, evaluated, and re-applied as new monitoring data and management questions emerge rather than treated as static decision-support tools.

In practice, stakeholders participate through structured decision-making processes such as advisory groups, technical committees, or public consultation. These structures shape how models are developed, interpreted, and applied, and operate across a range of decision contexts, from site-specific management to broader regional planning. Modelers may therefore interact directly with stakeholders in some adaptive management programs, while in others their contributions are incorporated more indirectly through these broader planning or regulatory processes. For modelers, critical methodological elements of model planning, development, evaluation and documentation (Schuwirth et al. 2019; Grimm et al. 2014) would therefore evolve alongside management processes. Clear communication of model assumptions, limitations, and updates is essential to ensure that outputs are appropriately interpreted and used in decision-making. Active collaboration is thus a crucial element of adaptive modeling that manifests in all aspects of model scoping, data and model sharing, and application. While this transdisciplinary and relational approach can have high transaction costs, it improves the relevance and salience of models (Addison et al. 2013), helps build trust (Herman et al. 2019), and increases accessibility for stakeholders (Langsdale et al. 2013).

In this paper, we explore the concept of adaptive modeling as an underutilized component of adaptive management. Our objectives are twofold: first, to describe how adaptive modeling can be integrated within the adaptive management cycle, and second, to discuss key challenges that limit its implementation and propose practical suggestions to address them. These challenges and opportunities were informed in part by discussions during a multidisciplinary workshop at the ‘Freshwater Down Under’ joint international meeting which brought together modelers, managers, and other stakeholders working at the science–management interface. We focus particularly on challenges related to communication, collaboration, and model utility.

Many of the challenges we identify have been documented in participatory modeling and transdisciplinary research (e.g., Gray et al. 2017, Voinov et al. 2018). For example, participatory modeling studies have highlighted difficulties in method selection, stakeholder engagement, and translating model outputs into management decisions as well as managing stakeholder expectations and balancing model complexity with usability (Zellner et al. 2022). More recent work has emphasized that barriers to the use of models in decision-making are not solely technical but also arise from communication and process-related factors including issues of trust, accessibility, and the context in which decisions are made (Weiskopf et al. 2022). Despite these advances, practical challenges

persist when integrating ecological models within iterative management processes, particularly linking model development to monitoring, evaluation, and decision-making over time.

Drawing on our experience in water resource management, we build on these insights to provide general guidance for strengthening the integration of adaptive modeling into adaptive management and monitoring frameworks. Our goal is to highlight how adaptive modeling can enhance adaptive management across diverse ecological subdisciplines, ecosystem types, and management needs, with a focus on applications in freshwater ecosystems where adaptive approaches are widespread. While adaptive management is often associated with site-level management decisions, the principles and potential actions discussed here can also inform broader regional planning processes.

Integrating Adaptive Modeling Into the Management Cycle

Our case study (Box 1, Irving et al. 2026) contrasts a commonly applied modeling approach in adaptive management (hereafter referred to as “traditional”) with our proposed integration of adaptive modeling. Adaptive management reduces uncertainty by iteratively learning from prior outcomes while developing hypotheses to improve future management and monitoring. Although adaptive management frameworks have evolved since their introduction in the 1970s (Holling 1978), the role of modeling within adaptive management has received less attention, despite its utility for informing management and monitoring (Conroy and Peterson 2013; Medema et al. 2008; Fischenich et al. 2019). The case study illustrates this gap: causal ecological models were not developed during the assessment phase (conceptual or quantitative), and although regular stakeholder meetings and monitoring occurred, the absence of an explicit, evidence-informed model early in the process limited the ability to interpret system response. Similar challenges have been reported in a range of environmental management applications, including environmental flow programs, species recovery planning, and watershed restoration, where models are often developed late in the management cycle or remain disconnected from monitoring design and decision-making processes (e.g., Peterson and Freeman 2016; Failing et al. 2013).

Adaptive modeling operates in connection with, and in parallel to, the adaptive management cycle (Fig. 1, Box 1). The adaptive management cycle shown here is conceptual and represents a generalized iterative process rather than a specific framework; however, it can support learning across

multiple levels of decision-making and adaptation (e.g., Pahl-Wostl 2009). Ideally, ecological models are integrated during the planning phase, where an explicit conceptual model is developed that depicts hypothesized cause-effect relationships based upon the best available evidence and expert knowledge. This model is used to define performance indicators, clarify management objectives and evaluation criteria, predict potential responses, aid in scenario development, and shape monitoring strategies (Gregory et al. 2006). In contrast to the traditional approach described above, an adaptive modeling framework can embed modeling from the start, aligning the conceptual model with management objectives. As monitoring is implemented during the management and monitoring phase, empirical models evolve through parameterization with incoming data and are then refined iteratively and evaluated to assess performance. In this way, monitoring serves to improve representation of system behavior over time. Monitoring also supports decision-making by assessing whether management-relevant thresholds (e.g., trigger values) are reached. Forecast-based triggers may also be used to initiate model updates or management responses when predicted conditions exceed predefined thresholds (e.g., Somers et al. 2018). Importantly, the relationship between monitoring and modeling is bidirectional. Monitoring data support model development, evaluation, and refinement, while model outputs can be used to identify key uncertainties, evaluate monitoring effectiveness, and guide adjustments to monitoring design (Dai et al., 2025). This iterative feedback helps to ensure that monitoring efforts remain aligned with management objectives, and fosters a dynamic connection between modeling and management that the traditional process in the case study lacked, even when thoughtfully constructed.

Key considerations for modelers and stakeholders in developing adaptive ecological models include clear, regularly revisited definitions for model structure, variables, parameterizations, and expected outcomes (Walters 1997). The proposed adaptive modeling approach explicitly emphasizes this early-stage alignment to ensure agreement on trigger values, indicators, and the communication of uncertainty. In contrast, during the evaluation phase in the traditional approach, trigger values were defined before any model development. This led to ambiguity later in the process, when models were introduced during evaluation but could not statistically support management thresholds. Open communication about model limitations and variability is crucial for managing stakeholder expectations and defining management thresholds that align with stakeholder risk tolerance. Iteratively clarifying a model's capabilities and limitations, while aligning outcomes and objectives

over time, strengthens trust and helps ensure the models remain decision-relevant throughout the adaptive management process.

Understanding and communicating the constraints on model development and application can be as crucial as defining modeling goals. Adaptive management cycles can vary in duration, from days (e.g., reservoir operations) to decades (e.g., a series of restoration projects executed over long timeframes), influencing the timing, resources, and expertise allocated to decisions. In the case study's traditional approach, models were introduced only after several years of flow reduction and monitoring, and only once sufficient data had accumulated. This delayed consideration reflects a shift in the "level of effort" (Harris et al. 2023), where professional judgment gave way to intensive modeling once justified by available data. However, the late-stage application of models combined with limited statistical power ultimately constrained the ability to update models during the adjustments phase. This highlights the merits of considering the "value of information" (VOI sensu Maxwell et al. 2015) early in the process so that models can be integrated early enough to guide decisions rather than to only evaluate them in hindsight.

A major test of a model lies in its integration within a broader monitoring and management framework (e.g., Fig. 1; Peterson and Freeman 2016). Models inform management decisions, while monitoring data evaluate the quality of both modeling and management strategies. Observed outcomes then serve to identify uncertainties in model parameters and predictions, update model structure, and formulate hypotheses to guide future management and monitoring (Box 1). Monitoring results are therefore used both to iteratively refine model structure and predictions (adaptive modeling) and to revise or adjust management interventions (adaptive management), which links learning directly to decision-making. However, monitoring data are often collected based on past sampling efforts rather than current management objectives (Walters 1997). Although the power analysis in the outcome phase of the case study indicated that increased sampling frequency would improve the ability to detect flow-related stress, the original monitoring plan remained unchanged in order to maintain consistency with past data collection efforts. An adaptive modeling approach would have enabled this analysis earlier, using uncertainty obtained from model parameter estimates to inform where to focus intensive monitoring efforts and to balance costs and benefits. This dynamic approach fosters effective decision-making, thereby improving resource management and conservation outcomes.

Box 1. Case study of integrating ecological models into an adaptive management plan (AMP) developed for Los Angeles County Sanitation Districts in 2018, based on Irving et al. (2026). The table contrasts how key stages of the adaptive management cycle unfolded in the case study and how they might have differed under an adaptive modeling framework. The AMP provided a monitoring strategy to evaluate potential ecological effects of reduced wastewater discharge from the Pomona, San Jose Creek, and Whittier Narrows Water Reclamation Plants (WRPs) to the San Gabriel River, California (USA). The monitoring plan includes a series of metrics identified as key habitat components necessary to maintain suitable riparian habitat for the federally threatened bird species, Least Bell's Vireo (*Vireo bellii pusillus*). The objective was to establish a level of flow reduction that would not impact habitat conditions to an unacceptable degree

Adaptive management section	Traditional adaptive management approach	Proposed adaptive modeling approach	Contrast between approaches
Assess Problem and Design	During the baseline monitoring period, regular updates and stakeholder meetings facilitated communication and alignment, enabling discussion of progress, challenges, and adjustments. A monitoring strategy was implemented and data were analyzed annually. No ecological model (conceptual or quantitative) was developed during this phase. A "trigger" value was defined to indicate when habitat conditions would be considered impacted beyond an acceptable level, however this did not follow a formal process.	Development of a conceptual model would occur during the planning phase alongside the adaptive management plan. This model would help define key stressors, indicators, and hypothesized system responses, informing the monitoring strategy and supporting a more robust selection of trigger thresholds.	In the traditional approach, monitoring design and trigger values were established prior to model development. This creates a disconnect between models, monitoring, and data. Under adaptive modeling, conceptual models would be developed during the planning phase, allowing monitoring indicators, trigger thresholds, and data collection strategies to be informed by model structure and anticipated uncertainty.
Management and Monitoring	Incremental flow reduction commenced after three years of baseline monitoring. Monitoring continued according to the established monitoring program.	Incremental flow reduction would still commence after the baseline monitoring period. However, as monitoring began, empirical model development would proceed alongside management actions, using incoming data to parameterize and refine models. Accessible and well-documented data management is essential to support this adaptive modeling process.	In the traditional approach, monitoring proceeds without a model guiding interpretation or identifying data needs. Under adaptive modeling, models are updated as monitoring data accumulate, allowing earlier assessment of system responses and identification of data gaps that may affect model performance.
Analyze and Evaluate	Models were developed and applied 2 years after flow reduction commenced using monitoring data collected during the program. Two ecological models were developed: an explanatory model to elucidate relationships in the data and a predictive model to anticipate future changes under different stress levels. A power analysis was conducted to determine the sampling frequency required to obtain statistically significant results. Models were validated, and variable importance and uncertainty were evaluated. Additional factors that could influence stress indicators (e.g., damage to trees from public activity) were included as variables.	Model development would begin much earlier and models would be iteratively updated as monitoring data accumulated. Model predictions would be regularly evaluated and re-parameterized as new data become available, allowing ongoing assessment of model performance.	In the traditional approach, models were developed several years into the monitoring program, meaning limitations in statistical power were identified only after substantial data had already been collected. Under adaptive modeling, earlier and iterative model development could reveal limitations in monitoring design sooner, allowing sampling strategies or model structure to be adjusted earlier in the adaptive management cycle.
Adjustments	Power analysis suggested increasing the frequency and duration of sampling. However, to retain consistency with the six years of previously collected data, the monitoring plan was continued without modification.	Model evaluation would inform adjustments to both model structure and monitoring design. If validation or predictive performance indicated limitations in the available data, modifications could include updating model parameters, revising model structure, or adjusting monitoring strategies (e.g., increasing sampling frequency or targeting specific variables).	In the traditional approach, limitations in statistical power were identified only after several years of monitoring, but monitoring design remained unchanged due to the desire to retain consistency with existing data. Under adaptive modeling, earlier evaluation of model performance could highlight monitoring limitations sooner, allowing sampling strategies or data collection priorities to evolve earlier in the adaptive management process.
Outcome	Models were unable to detect a statistically significant relationship between reduced flow and tree stress; consequently, explicit trigger thresholds could not be defined. Power analysis suggested that approximately ten additional years of monitoring would be required to detect the effect of flow reduction on habitat with sufficient statistical confidence.	Earlier and iterative model evaluation could reveal limitations in predictive performance and monitoring sufficiency sooner in the process. This feedback would allow model structure, variables, or monitoring strategies to be adjusted during the management program, improving the ability to evaluate relationships between management actions and ecological responses over time.	In the traditional approach, the inability to detect statistically significant relationships was only identified after several years of monitoring and model development, leaving little opportunity to modify monitoring design. Under adaptive modeling, earlier feedback from models could highlight data limitations and guide adjustments to monitoring or model structure sooner. While uncertainty would still remain, the adaptive process can increase the likelihood that management-relevant thresholds and indicators are identified.

Facilitating Communication and Relationships

Insights from our multi-disciplinary workshop highlighted several key elements required to effectively integrate adaptive modeling into decision-making processes. The following discussion draws on these experiences to outline the relational and procedural factors that most strongly influence the success of adaptive modeling in practice.

Effective collaboration is central to an integrated adaptive modeling approach and is increasingly framed as the co-production of knowledge, in which modelers and stakeholders jointly contribute expertise, values, and contextual understanding to develop shared decision-relevant outputs. Within adaptive modeling, this collaborative process is not ancillary but directly shapes model structure, assumptions, and ultimately its uptake in decision-making contexts.

While co-production is often conceptualized as direct engagement between modelers and stakeholders, in many applied contexts these interactions are mediated through institutional processes such as land-use planning, environmental impact assessment, or advisory structures. Modelers may contribute indirectly in such cases, with communication and interpretation of model outputs occurring through intermediaries. The considerations outlined here apply across both direct and indirect engagement settings, as the effectiveness of adaptive modeling depends on how well model assumptions, uncertainties, and outputs are communicated and integrated into decision-making processes.

Transparency supports this process by enabling clear communication about model assumptions, uncertainties, and limitations. It also underpins trust, which influences how model outputs are interpreted and used in decision-making. Trust has been shown to depend on perceptions of logic (competence), authenticity, and empathy (Frei and Morriss 2020). These dimensions are particularly relevant in adaptive modeling, where iterative feedback between stakeholders, intermediaries, and modelers requires ongoing negotiation of expectations, assumptions, and trade-offs.

Differences in objectives between modelers and stakeholders can affect how models are developed and applied. Stakeholders may prioritize robust, defensible decision support, whereas modelers may focus on methodological innovation or technical advancement. Making these motivations explicit is therefore important for aligning model development with decision needs. This alignment is a core feature of co-produced modeling processes and directly influences whether model outputs are considered credible, salient, and usable.

Challenges arise when stakeholders are reluctant to articulate their needs, or when model development does not

sufficiently engage with the broader decision context. In such cases, models may fail to capture key aspects of the system, including governance constraints, implementation pathways, and monitoring requirements (Kingsford et al. 2017). Similarly, insufficient communication of model uncertainty can limit the usefulness of outputs for decision-making under risk (Rumpff et al. 2011).

Empathy and mutual understanding can facilitate early clarification of goals, assumptions, and acceptable levels of uncertainty, improving the relevance and usability of model outputs (Scoggins et al. 2022). In this way, social dynamics such as trust and communication are not peripheral considerations but influence the effectiveness of adaptive modeling as a decision-support tool, particularly in iterative and uncertain management contexts.

This perspective complements existing participatory and iterative modeling approaches. For example, some frameworks emphasize structured iteration between stakeholder workshops (Smajgl 2010) or broader participatory modeling objectives (Gray et al. 2018). Others highlight relational dimensions of knowledge co-production, where modeling is shaped by social context, values, and power relations (e.g., Klein et al. 2024) or constraints such as data availability and agreement on management goals within decision contexts (Schlüter et al. 2025).

We do not focus here on specific participatory methods or workshop structure but rather on how these social and contextual factors influence the integration and iterative use of ecological models within adaptive management cycles. In particular, we emphasize how collaboration and communication shape model development, and the uptake of model outputs in decision-making.

In our case study, structured engagement through regular meetings supported ongoing alignment between modelers and stakeholders. However, the absence of an ecological model early in the process limited opportunities for stakeholders to engage with, critique, and refine model assumptions. Earlier integration of modeling could have strengthened this co-production process by providing a shared framework for exploring system dynamics, uncertainties, and potential outcomes, thereby enhancing both trust and decision relevance.

Collaborative Model Development

Successful adaptive management requires sustained engagement and buy-in by a wide range of stakeholders (Webb et al. 2018; Fischenich et al. 2019). Adaptive modeling is no different. Here, we identify leverage points for collaboration in adaptive modeling; these are based on our experiences as modelers and informed by the literature on collaborative or stakeholder mediated modeling (e.g.,

van den Belt 2004; Grant and Swannack 2008; Langsdale et al. 2013; Gray et al. 2017; Herman et al. 2019).

Early and sustained stakeholder involvement in model planning can balance competing objectives, fostering the creation of purpose-specific models that make reliable predictions and are readily adopted by both modelers and stakeholders alike. Ideally, collaboration would begin at the project's outset, with active listening to ensure the model's purpose is clear and to establish shared objectives (Cartwright et al. 2016). These discussions are best held in environments conducive to authentic conversations and often benefit from expert facilitation. Models developed through effective collaboration within a shared conceptual framework are more likely to be considered as legitimate by all stakeholders (Langsdale et al. 2013; Mussehl et al. 2023). In our case study, the delayed development of ecological models limited opportunities for stakeholders to engage with the modeling process early on, potentially affecting the alignment between model outputs and stakeholder expectations.

Adaptive modeling benefits from direct interaction between modelers and stakeholders during problem scoping. Jointly developed conceptual models enhance legitimacy and ensure numerical models are fit for the intended purpose (Herman et al. 2019). Understanding management targets is crucial as they influence model selection. For example, trajectory-based objectives (e.g., 'maintain or improve' ecological condition), may require less sophisticated models than those with specific quantitative objectives (e.g., X% increase in the biomass of Y within 10 years). As quantitative tools become more sophisticated, stakeholders may play a smaller role but still contribute to the process by providing data sets or expert knowledge (de Little et al. 2018; Rowley et al. 2020). During evaluation, stakeholders can help assess model predictions relative to their lived experiences and by reviewing the models' usability (see Box 1).

Early model implementation may reveal poor predictive ability, but this is an opportunity for improvement rather than a model failure. Conceptual models, model structure, parameters, or monitoring can be adjusted during the 'analyze and evaluate' and 'adjustment' phases of the adaptive management cycle to improve model performance. However, it is important to note the different temporal scales of adjustments for management, modeling and monitoring. While management actions may be revised less frequently, monitoring and modeling can iterate on shorter cycles, allowing new information to be incorporated and evaluated as it becomes available. Decisions to update models are best made collaboratively, considering explained variation, prediction accuracy, uncertainty, usability, and available resources.

Stakeholders can identify and fill important data gaps during model development. Differences in data collection

priorities and the value placed on certain variables can be explored by comparing competing model structures (e.g., Williams et al. 1996). Using model selection approaches to test various combinations and parameterizations helps stakeholders determine which variables yield the greatest explanatory power (Johnson and Omland 2004), thereby guiding more effective data collection and monitoring efforts. In our case study, the data collection strategy was thoughtfully designed and implemented over several years. However, when ecological models were later developed and applied, it was evident that existing data were insufficient to establish meaningful statistical relationships. This experience underscores the importance of integrating modeling considerations early in the adaptive management process to ensure that data collection efforts align with modeling requirements.

Regular feedback between modelers and stakeholders on monitoring strategy, model structure, and outcomes helps to adaptively align these elements with shifting community values, environmental conditions, and management priorities (Pahl-Wostl 2007; Ringold et al. 1996). While dialog and relationship-building with stakeholders can be resource and time-intensive, the investment can pay off with improved buy-in, trust, and decision salience.

To synthesize these practical considerations, we outline key benefits and risks associated with integrating adaptive modeling within adaptive management frameworks (Table 1).

Creating Accessible Model Workflows

Adaptive modeling requires access to datasets and model code along with well-reported and transparent workflows. These "open science" methods (Murray-Rust, 2008) are particularly important in the context of adaptive modeling, where models may be developed, applied, and refined over extended periods or by different modelers. It is wise to incorporate transparent data management early in the 'assess problem and design' phase to facilitate adaptability and best practices (Box 1).

Data access and data wrangling are key activities in ecological modeling, and modern repositories like GitHub, Zenodo, and Dryad provide archives for data, code, and documentation, often with a digital object identifier (DOI). Proper archival practices are especially important for novel and historical datasets, allowing them to be digitally stored with models. We also encourage efforts to compile and centralize "non-traditional" data, such as First Nations science and knowledge (Scales et al., 2013). Best practices for these data are developing, but ensuring free, prior, and informed consent (FPIC per the United Nations Declaration on the Rights of Indigenous Peoples 2007) and prioritizing

Table 1 Risks and Benefits of integrating adaptive modeling within adaptive management frameworks

Category	Description	Example
BENEFIT	Improved ability to update predictions as monitoring data are collected and the model is updated.	Nichols et al. (2007)
	Improved stakeholder understanding of system function and opportunity for two-way learning through the model update process.	Horne et al. (2022)
	Early integration improves monitoring by focusing on variables needed for model evaluation.	Dai et al. (2025), Failing et al. (2013)
	Supports iterative refinement of management strategies and program design over time.	Littles et al. (2022)
RISK	Too much focus on the technical side of management may reduce stakeholder engagement, which is essential for long-term success of adaptive management goals.	Webb et al. (2018)
	Time lags between implementation, monitoring, and ecological response can delay learning and decision-making.	Littles et al. (2022)
	Decisions remain subjective due to competing trade-offs.	Failing et al. (2013)

CARE principles for Indigenous data (Collective benefit, Authority to control, Responsibility and Ethics per Carol et al. 2020) provide a respectful framework for collaboration that protects Indigenous Cultural Intellectual Property (ICIP). Subsequent integration of First Nations science into the adaptive modeling cycle is a co-development process, and the nature of implementation will vary in response to the specific cultural, ecological, and relational contexts negotiated between First Nations and research partners.

Archiving model code enhances accessibility and transparency, thus supporting peer review (Barnes 2010) and long-term adjustment in adaptive management. Models can be archived publicly or privately in different repositories (e.g., Dryad vs. GitHub) based on context-specific need for privacy and collaboration. Interactive documentation (e.g., Markdown or Jupyter notebooks) supports adaptive modeling and ensures continuity despite changes in the modeling team. Thus, the most effective model structures are stable, easy to implement, and have clear annotations, including informative error messages.

Model outputs are reported during the 'analyze and evaluate' phase of adaptive modeling (Fig. 1), creating a benchmark for model iteration and providing feedback to stakeholders. While reporting standards exist for some models (e.g., species distribution models; Zurell et al. 2020), all reports would, at a minimum, benefit from a detailed description of the model design and testing, related publications, references to earlier iterations, and links to data and code repositories. Reports could also offer clear recommendations for adapting monitoring and model design.

While robust archiving and reporting allow proficient users to interrogate the model, not all stakeholders have the necessary technical skills to perform this task. User interfaces (e.g., the R Shiny App or USEPA's Freshwater Explorer) help bridge this gap by presenting complex model outputs in a user friendly and intuitive format. Although these interfaces require updates as programming languages

or models evolve, they can dramatically improve stakeholder engagement and accessibility.

Summary

Our paper synthesizes key challenges and provides suggestions to improve the integration of adaptive modeling into adaptive management for diverse stakeholder groups. Key challenges include effective stakeholder collaboration and clear communication of model limitations along with model transparency. By addressing these challenges and fostering collaboration, adaptive modeling can improve adaptive management, ensuring models align with decision-making needs.

The following suggestions provide a starting point for modelers to operationalize the adaptive modeling process outlined here:

1. Commit to building trust by valuing authenticity, logic, and empathy in all communication.
2. Collaborate early with stakeholders to co-create a shared vision, clear objectives, and an agreed upon conceptual model.
3. Communicate uncertainty to stakeholders to find ways to improve model strategies.
4. Ensure transparent data acquisition, modeling process, and archiving to support future model adaptations.
5. Prioritize model accessibility, regardless of skill level. This requires archiving, visualization, and post-processing of outcomes along with a communication strategy.
6. Maximize the use of adaptive modeling with stakeholders: while it is likely to be resource and time-intensive, the return-on-investment can be high.

Through strategies like these, we believe that adaptive modeling can improve adaptive management processes and

outcomes to the benefit of all stakeholders and the environments they are seeking to protect.

Data availability

No datasets were generated or analysed during the current study.

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Compliance with ethical standards

Competing Interests The authors declare no competing interests.

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